

Abstract

AI-Enabled Adaptive Silent V2X Alert System for Traffic Noise Reduction is a smart transportation safety system developed to address two major problems in modern urban environments: road accidents and increasing traffic noise pollution. Conventional traffic safety mechanisms heavily depend on vehicle horns, sirens, and audible warning systems to alert drivers and pedestrians. Although these methods can provide immediate warnings, excessive and unnecessary use of sound contributes to noise pollution, driver distraction, and discomfort in densely populated areas. The proposed system introduces an intelligent, communication-based approach that provides safety alerts while minimizing the dependence on conventional audible signals. The proposed system utilizes Artificial Intelligence (AI), Vehicle-to-Everything (V2X) communication, Internet of Things (IoT), sensors, and adaptive alert technologies to enable real-time communication between vehicles, roadside infrastructure, pedestrians, and other connected road users. Vehicles can exchange important information such as speed, location, direction, distance, braking conditions, and potential hazards. The collected information is processed by an AI-based decision-making module to identify dangerous situations and estimate the level of risk. Based on the detected risk, the system generates an appropriate alert using a multi-level adaptive warning mechanism. For low-risk situations, the driver can receive silent notifications through a dashboard display or mobile application. For moderate-risk situations, the system can provide visual and vibration-based alerts. In high-risk or emergency situations, an audible warning can be activated when necessary. This adaptive mechanism avoids unnecessary horns and alarms while ensuring that critical warnings are not missed. The system can also incorporate GPS, cameras, ultrasonic sensors, proximity sensors, microphones, and environmental sensors to improve situational awareness. AI-based algorithms can analyze traffic patterns, detect nearby obstacles, identify abnormal vehicle behavior, and predict possible collision scenarios. V2X communication allows this information to be shared with nearby connected vehicles and roadside units, enabling drivers to receive warnings before they can visually identify a potential hazard. Another important feature of the proposed system is its potential application in smart cities and intelligent transportation systems (ITS). Traffic authorities can use roadside communication units to broadcast information about accidents, congestion, road construction, emergency vehicles, school zones, pedestrian crossings, and other hazardous conditions. Connected vehicles can receive these alerts and automatically provide appropriate warnings to drivers without requiring continuous use of horns or sirens.

Keywords: Artificial Intelligence (AI), Vehicle-to-Everything (V2X), Intelligent Transportation System (ITS), Traffic Noise Reduction, Silent Alerts, Adaptive Alert System.